Carnegie Mellon

School of Computer Science

Deep Reinforcement Learning and Control

Multi-goal RL and IL

Spring 2025, CMU 10-403

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Imitation learning for diverse goals

- Pushing to diverse locations
- Pouring to different bottles
- Driving to different destinations

We need a way to communicate the goal during learning of the policy

Multi-goal Imitation learning and RL

- Often times we care to learn policies that achieve many related goals
- For example: push object A to (10,10,10) and to (10,12,10)
- The two policies should have many things in common
- Training such policies jointly may be beneficial

Universal value function approximators

$$V(s;\theta) \rightarrow V(s,g;\theta)$$

$$\pi(s;\theta) \rightarrow \pi(s,g;\theta) \quad s,g \in \mathcal{S}$$

- The experience tuples should contain the goal.

$$(s,a,r,s') \rightarrow \left(s,g,a,r,s'\right)$$

Universal Value Function Approximators, Schaul et al.

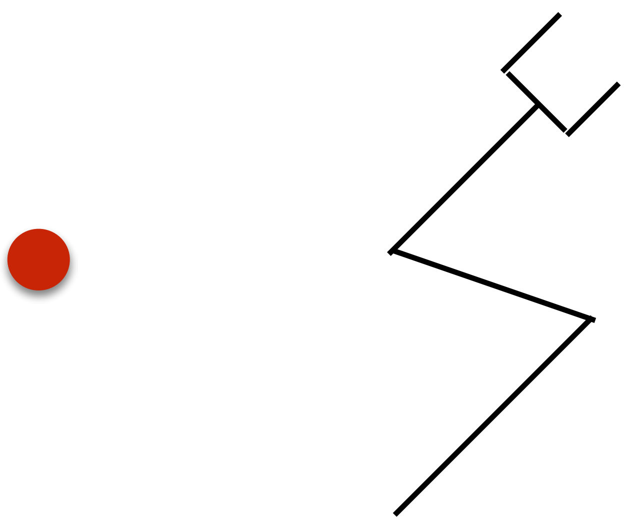
Goal conditioned behavior cloning

- Assumes access to a set of trajectories $\mathcal{T} = \{o_1^j, a_1^j, o_2^j, a_2^j, o_3^j, a_3^j, \ldots, o_T^j, a_T^j, g^j, j = 1...T\}$. Trains a policy by minimizing a standard supervised learning objective:

$$\mathcal{L}_{BC}(\theta, \mathcal{T}) = \mathbb{E}_{(s_t^j, a_t^j, g^j) \sim \mathcal{T}} \left[\|a_t^j - \pi_\theta(s_t^j, g^j)\|_2^2 \right]$$

Goal relabelling

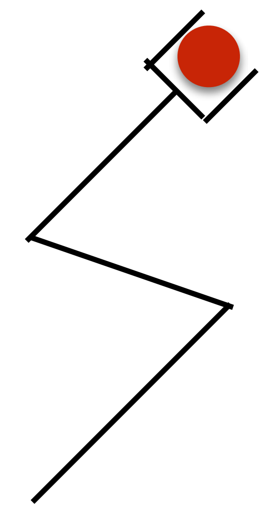
Idea: use failed executions under one goal g, as successful executions under an alternative goal g' (which is where we ended at the end of the episode).



No reward :-(

Goal g

Our reacher at the end of the episode $(s, g, a, 0, s')$



Goal g'

reward :-)

Our reacher at the end of the episode $(s, g', a, 1, s')$

Hindsight Experience Replay

Marcin Andrychowicz*, **Filip Wolski**, **Alex Ray**, **Jonas Schneider**, **Rachel Fong**,
Peter Welinder, **Bob McGrew**, **Josh Tobin**, **Pieter Abbeel**†, **Wojciech Zaremba**†
OpenAI

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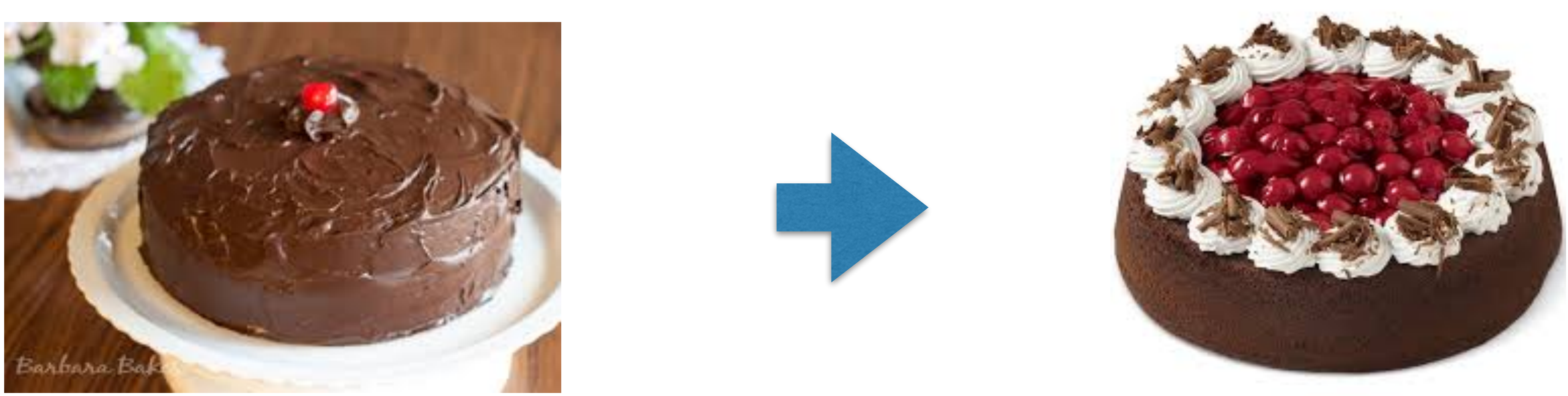


RL with goal relabelling

Algorithm 1 Hindsight Experience Replay (HER)

Given:

- an off-policy RL algorithm $\mathbb{A}$, ▷ e.g. DQN, DDPG, NAF, SDQN
- a strategy $\mathbb{S}$ for sampling goals for replay, ▷ e.g. $\mathbb{S}(s_0, \ldots, s_T) = m(s_T)$
- a reward function $r : \mathcal{S} \times \mathcal{A} \times \mathcal{G} \to \mathbb{R}$. ▷ e.g. $r(s, a, g) = -[f_g(s) = 0]$

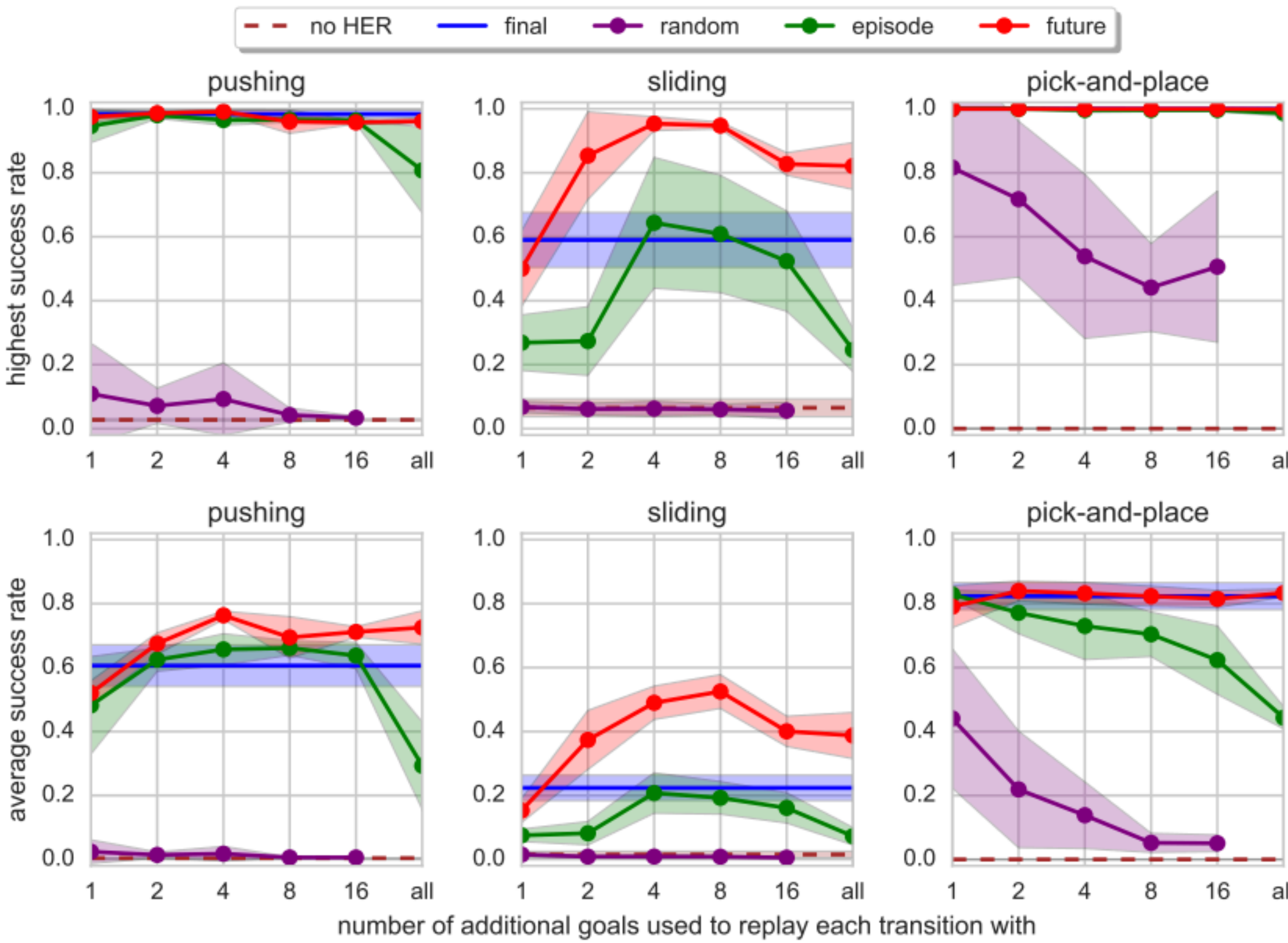
Initialize $\mathbb{A}$ ▷ e.g. initialize neural networks
Initialize replay buffer R
for episode $= 1, M$ **do**
  Sample a goal g and an initial state s_0.
  for $t = 0, T-1$ **do**
    Sample an action a_t using the behavioral policy from $\mathbb{A}$:
      $a_t \leftarrow \pi_b(s_t||g)$ ▷ || denotes concatenation
    Execute the action a_t and observe a new state s_{t+1}
  end for
  for $t = 0, T-1$ **do**
    $r_t := r(s_t, a_t, g)$
    Store the transition $(s_t||g, a_t, r_t, s_{t+1}||g)$ in R ▷ standard experience replay
    Sample a set of additional goals for replay $G := \mathbb{S}(\textbf{current episode})$
    for $g' \in G$ **do**
      $r' := r(s_t, a_t, g')$
      Store the transition $(s_t||g', a_t, r', s_{t+1}||g')$ in R ▷ HER
    end for
  end for
  for $t = 1, N$ **do**
    Sample a minibatch B from the replay buffer R
    Perform one step of optimization using $\mathbb{A}$ and minibatch B
  end for
end for

The reward here is $||s_t - g||$

G : the states of the current episode

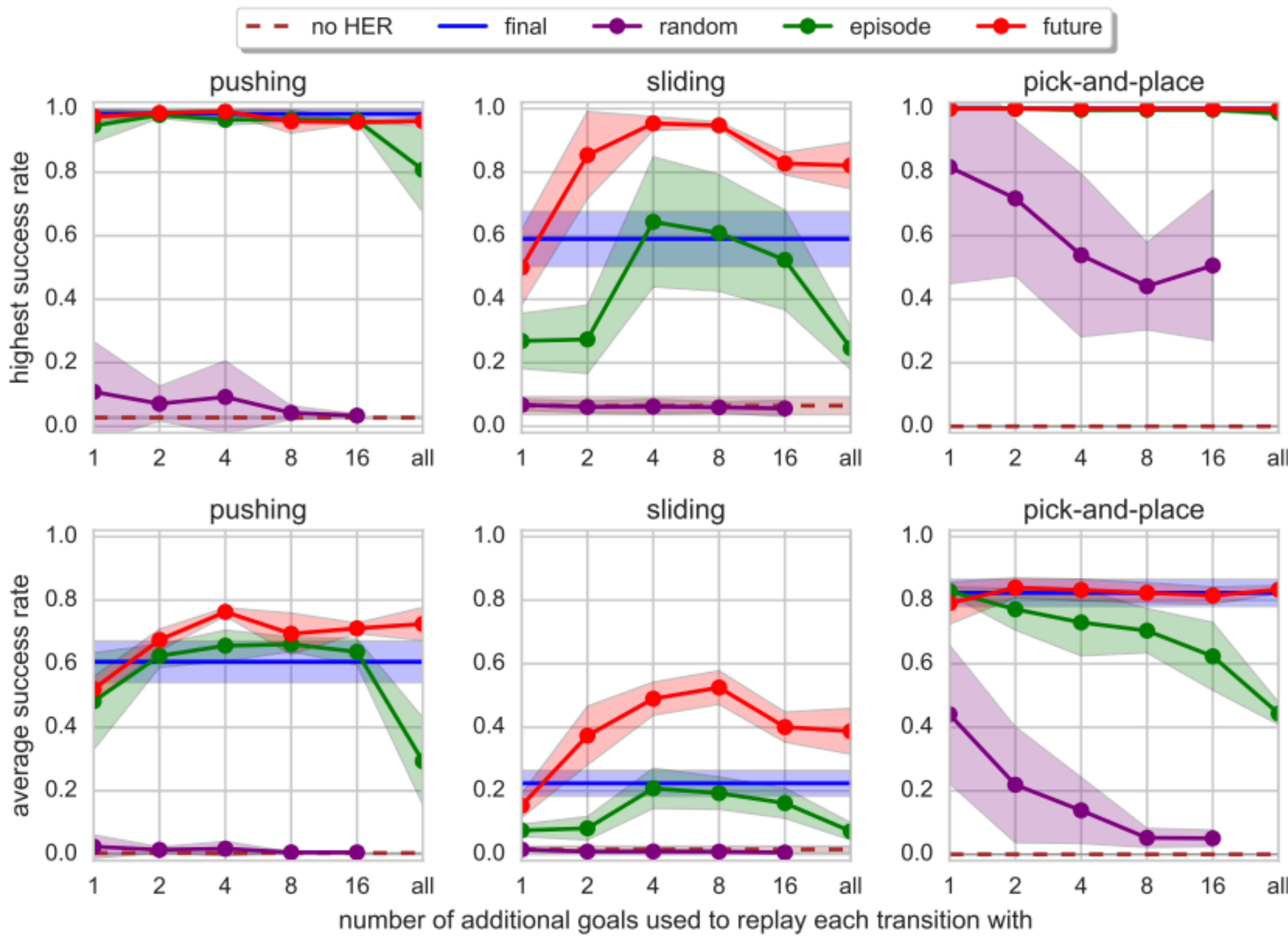
Usually as additional goal we pick the goal that this episode achieved, and the reward becomes non zero

How to select states for relabelling



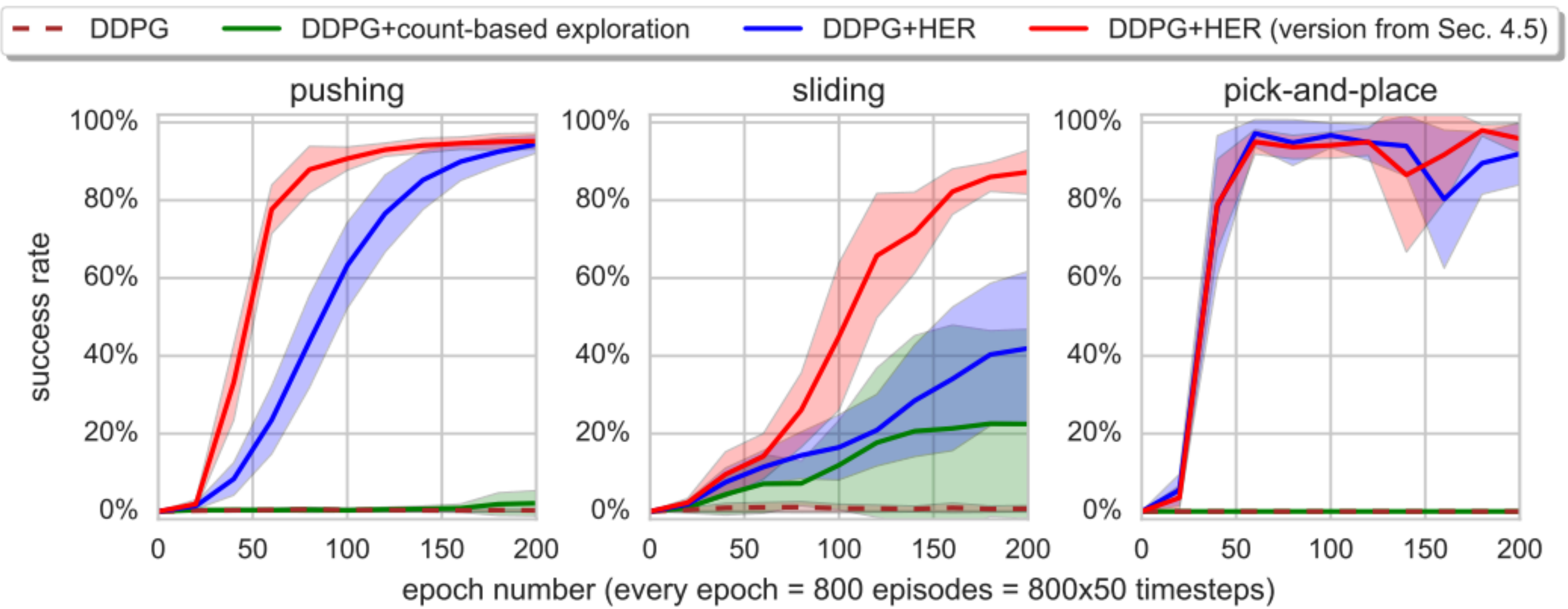
- Final: for each state, use the last state reached in the episode as a goal
- Future: for each state, use random 4 states (observed after the state) in the same episode as a goal

How to select states for relabelling



Above 8, the relabelled data are way more than real data, performance degrades.

How to select states for relabelling



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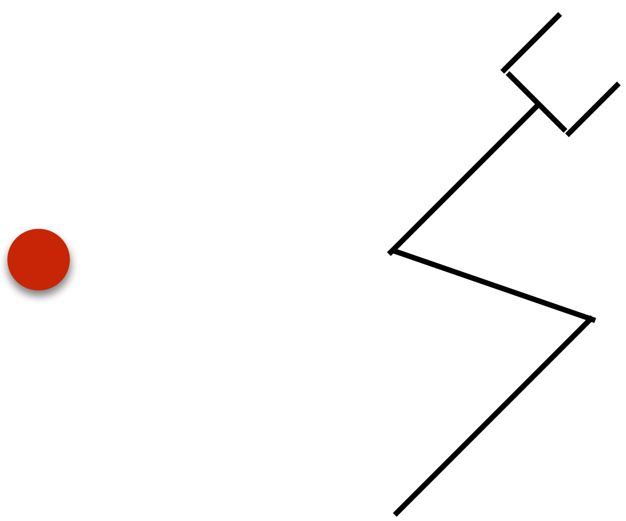

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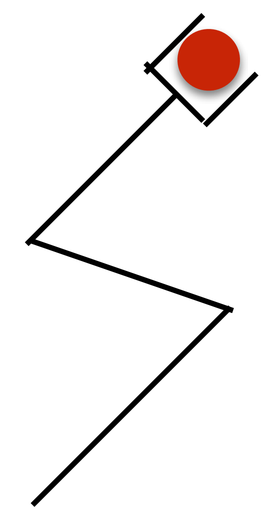

Goal g'

reward :-)

Our reacher at the end of the episode $(s, g', a, 1, s')$

In which of the tasks you think goal relabelling will help the most?

- Picking an object up?
- Reaching?
- Pushing an object to a location?
- All of the above considered jointly?

Goal-conditioned Imitation Learning

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Three ideas:

- Goal-conditioned GAIL
- Combining RL and imitation rewards
- Goal relabelling in both agent and expert trajectories

Goal relabelling expert trajectories

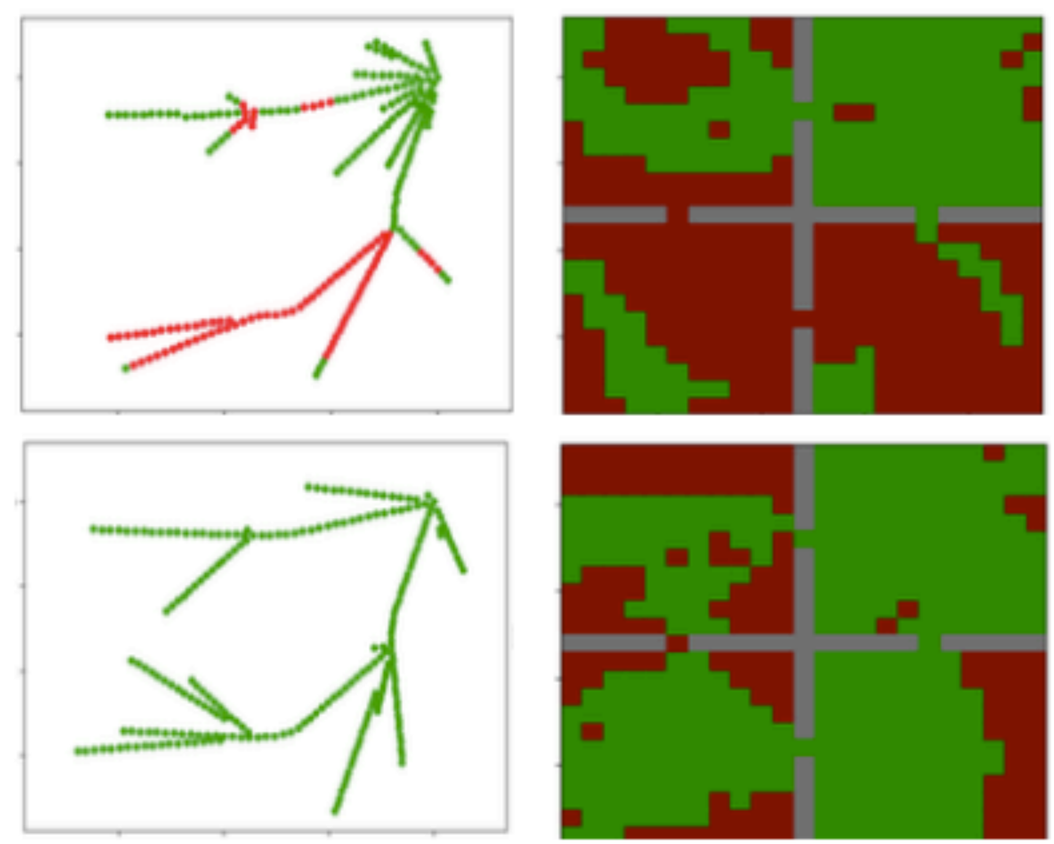
If $(s_t^j, a_t^j, s_{t+1}^j, g^j)$ is in a demonstration trajectory, we also add $(s_t^j, a_t^j, s_{t+1}^j, g' = s_{t+k}^j)$

Data augmentation via goal relabelling on demonstrations!

BC

Green means the policy visited these goals

BC with goal relabelling



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Data augmentation via goal relabelling on demonstrations!

Relabelling can be used in GCBC, GCBC+HER, GoalGAIL,

BC

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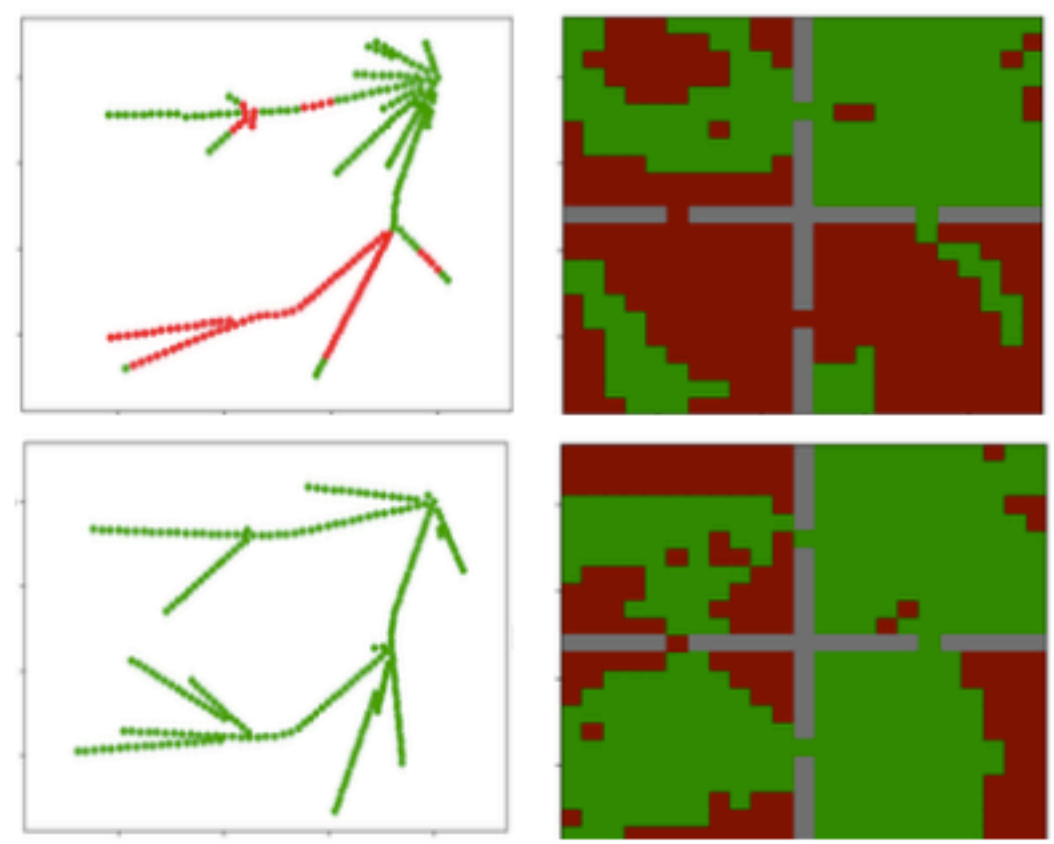
BC with goal relabelling



Goal GAIL

Input: Expert trajectories , initial policy parameters θ_0 and initial discriminator weights ϕ_0.

For i=0,1,2,3… **do**

1. Sample agent trajectories $\tau_i \sim \pi_{\theta_i}$ (for each trajectory, we sample a goal and then run the goal conditioned policy in the environment)
2. Update the discriminator parameters with the gradient:

$$\mathbb{E}_{(s,a,g)\sim\text{Demo}}[\nabla_\phi \log(1 - D_\phi(s,a,g))] + \mathbb{E}_{(s,a,g)\in\tau_i}[\nabla_\phi \log(D_\phi(s,a,g))]$$

3. Update the policy using a policy gradient computed with the rewards, e.g., the REINFORCE policy gradient would be:

$$\mathbb{E}_{(s,a,g)\in\tau_i}[\nabla_\theta \log \pi_\theta \log D_{\phi_{i+1}}(s,a,g)]$$

end for

Combining imitation and task rewards

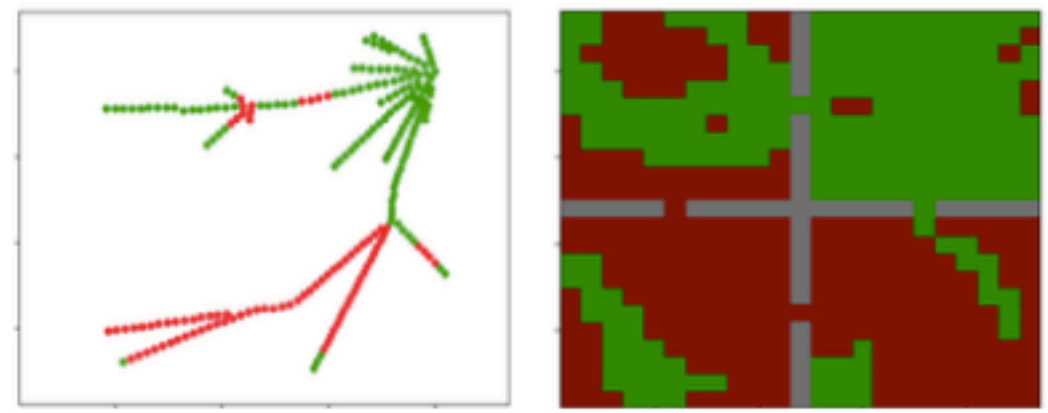
$$r(s, a) = \lambda r_{GAIL}(s, a) + (1 - \lambda) r_{task}(s, a), \quad \lambda \in [0,1] \,.$$

Compute Q functions and update the policy using DDPG:

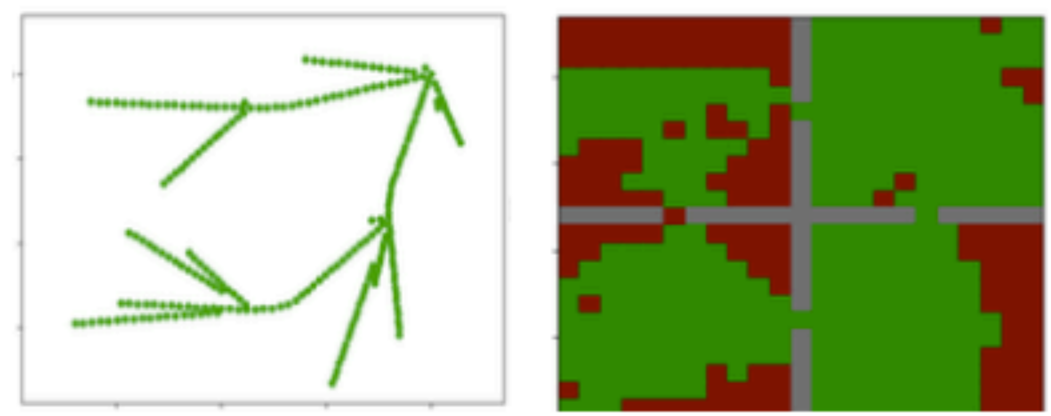
$$\nabla_\theta \hat{J} = \frac{1}{N} \sum_{i=1}^{N} \nabla_a Q_\phi(a, s, g) \nabla_\theta \pi_\theta(s, g)$$

Goal conditioned imitation learning

BC

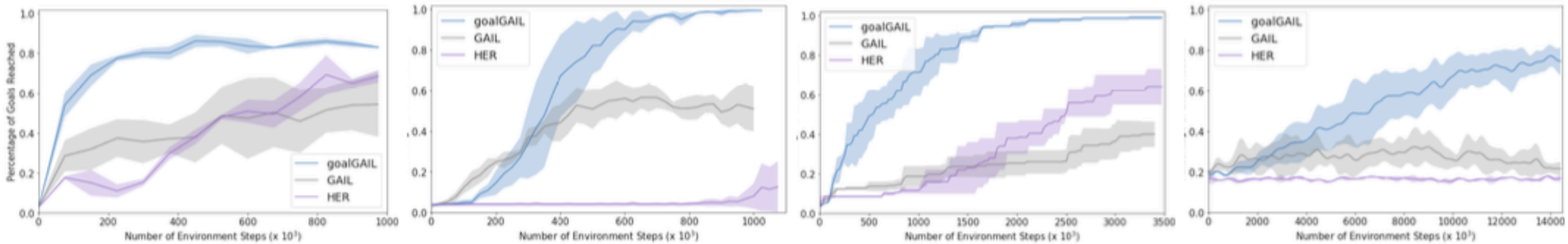


BC with goal relabelling

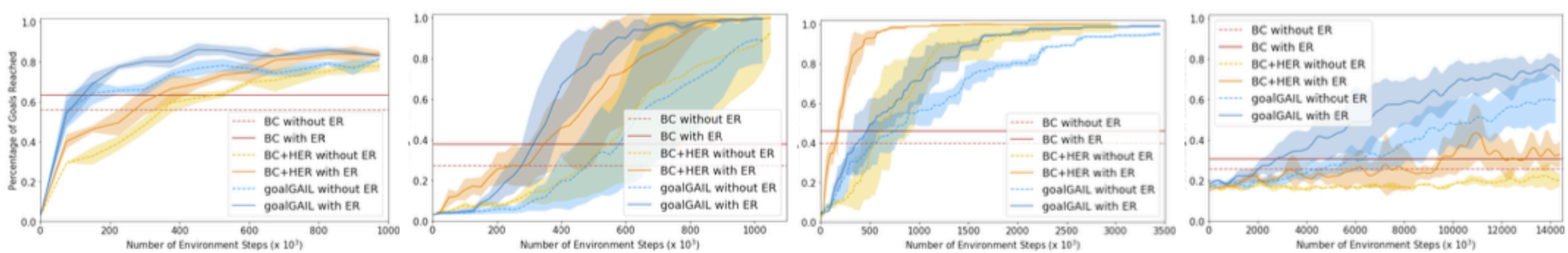


Algorithm 1 Goal-conditioned GAIL with Hindsight: *goalGAIL*

Input: Demonstrations $\mathcal{D} = \{(s_0^j, a_0^j, s_1^j, ..., g^j)\}_{j=0}^{D}$, replay buffer $\mathcal{R} = \{\}$, policy $\pi_\theta(s, g)$, discount γ, hindsight probability p
while not done **do**
    # Sample rollout
    $g \sim \texttt{Uniform}(\mathcal{S})$
    $\mathcal{R} \leftarrow \mathcal{R} \cup (s_0, a_0, s_1, ...)$ sampled using $\pi(\cdot, g)$
    # Sample from expert buffer and replay buffer
    $\{(s_t^j, a_t^j, s_{t+1}^j, g^j)\} \sim \mathcal{D}$, $\{(s_t^i, a_t^i, s_{t+1}^i, g^i)\} \sim \mathcal{R}$
    # Relabel agent transitions
    for each i, with probability p **do**
        $g^i \leftarrow s_{t+k}^i, \quad k \sim \text{Unif}\{t+1, \ldots, T^i\}$ ▷ Use *future* HER strategy
    end for
    # Relabel expert transitions
    $g^j \leftarrow s_{t+k'}^j, \quad k' \sim \text{Unif}\{t+1, \ldots, T^j\}$
    $r_t^h = \mathbb{1}\left[s_{t+1}^h == g^h\right]$
    $\psi \leftarrow \min_\psi \mathcal{L}_{GAIL}(D_\psi, \mathcal{D}, \mathcal{R})$ (Eq. 3)
    $r_t^h = (1 - \delta_{GAIL}) r_t^h + \delta_{GAIL} \log D_\psi(a_t^h, s_t^h, g^h)$ ▷ Add annealed GAIL reward
    # Fit Q_ϕ
    $y_t^h = r_t^h + \gamma Q_\phi(\pi(s_{t+1}^h, g^h), s_{t+1}^h, g^h)$ ▷ Use target networks $Q_{\phi'}$ for stability
    $\phi \leftarrow \min_\phi \sum_h \|Q_\phi(a_t^h, s_t^h, g^h) - y_t^h\|$
    # Update Policy
    $\theta += \alpha \nabla_\theta \hat{J}$ (Eq. 2)
    Anneal δ_{GAIL} ▷ Ensures outperforming the expert
end while



(a) Continuous Four rooms (b) Pointmass block pusher (c) Fetch Pick & Place (d) Fetch Stack Two



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